Workforce Attrition Patterns and Risk Hotspot Analysis

Palo Alto Networks

Diagnostic study of where, among whom, and under what conditions employees leave

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Abstract

This study examines workforce attrition at Palo Alto Networks using records for 1,470 employees, of whom 237 left (an overall attrition rate of 16.1%). Rather than predicting individual exits, it diagnoses where attrition is concentrated so that HR action can be targeted instead of generalized. Attrition is highly uneven. Employees who work overtime leave at 30.5% versus 10.4% for those who do not, frequent business travelers at 24.9% versus 8.0% for non-travelers, and employees in their first two years at 29.8%. Sales Representatives (39.8%), Laboratory Technicians (23.9%) and the Sales department (20.6%) are the clearest functional hotspots. Gender, education level and years since last promotion show no statistically significant association with attrition. In a multivariable model, overtime, frequent travel and the absence of stock options remain strongly associated with exit after controlling for age, job level and tenure. The paper closes with evidence-based retention priorities and a Streamlit dashboard for continued monitoring.

Keywords: employee attrition, workforce analytics, overtime, business travel, tenure, retention, HR dashboard

1. Introduction

In a high-skill, fast-paced cybersecurity organization, attrition is costly. It removes critical technical expertise, raises hiring and onboarding costs, disrupts long-running projects and can depress the morale and productivity of the people who stay. Before leadership can predict attrition or design retention programs, it must first understand where attrition is happening and which groups are most affected. This project provides that foundation.

The analysis answers four questions posed by HR leadership:

- Which departments and job roles experience the highest attrition?
- Is attrition concentrated among specific age groups or tenure bands?
- Do workload factors such as overtime and business travel contribute to attrition?
- Are early-career employees leaving more frequently than experienced ones?

2. Data and Methodology

2.1 Dataset

The dataset contains 1,470 employee records with 31 fields covering demographics, job characteristics, compensation, satisfaction ratings and tenure. The outcome variable, *Attrition*, is coded 1 if the employee left and 0 if retained. Validation found no missing values, no duplicate records and no inconsistent attrition labels, so no records were removed. Text categories (department, job role, travel frequency) were standardized to consistent labels.

2.2 Analytical approach

The analysis followed six steps: (1) data validation and cleaning; (2) overall attrition assessment to establish the baseline; (3) department and role-wise analysis; (4) demographic analysis by age, gender, marital status and education; (5) tenure and career-stage analysis, including promotion stagnation; and (6) workload and mobility analysis covering overtime, travel and commute distance. Continuous variables were grouped into bands (for example age 18-24, 25-34, 35-44, 45-54, 55+; tenure 0-1, 2-3, 4-5, 6-10, 11+ years).

Group differences were tested with Pearson chi-square tests of independence (significance threshold 0.05). To check whether workload effects survive adjustment for who is exposed to them, a multivariable logistic regression was fitted with attrition as the outcome. A segment is flagged as a **risk hotspot** when its attrition rate is at least 1.25 times the organizational baseline (16.1%) and it contains at least 30 employees.

2.3 Key performance indicators

KPI	Definition	Value
Attrition rate	Employees who left / total employees	16.1%
Department attrition rate	Attrition within each department	Sales 20.6%, HR 19.0%, R&D 13.8%
Role attrition rate	Attrition within each job role	Range 2.5% to 39.8%
Early-tenure attrition	Attrition among employees with 2 or fewer years at the company	29.8%
Workload Attrition Index	Attrition rate of employees who work overtime OR travel frequently, divided by the rate of all other employees	2.95x

Note: the Workload Attrition Index is defined here by the author because the project brief does not specify a formula. A value above 1 means workload-exposed employees leave more often.

3. Results

3.1 Overall attrition and department view

Of 1,470 employees, 237 left, giving a baseline attrition rate of 16.1% (about one in six employees). Sales has the highest department attrition at 20.6%, followed by Human Resources at 19.0% (a small group of 63) and Research & Development at 13.8%. The department differences are statistically significant (chi-square $p = 0.005$), but they are modest compared with the differences between roles.

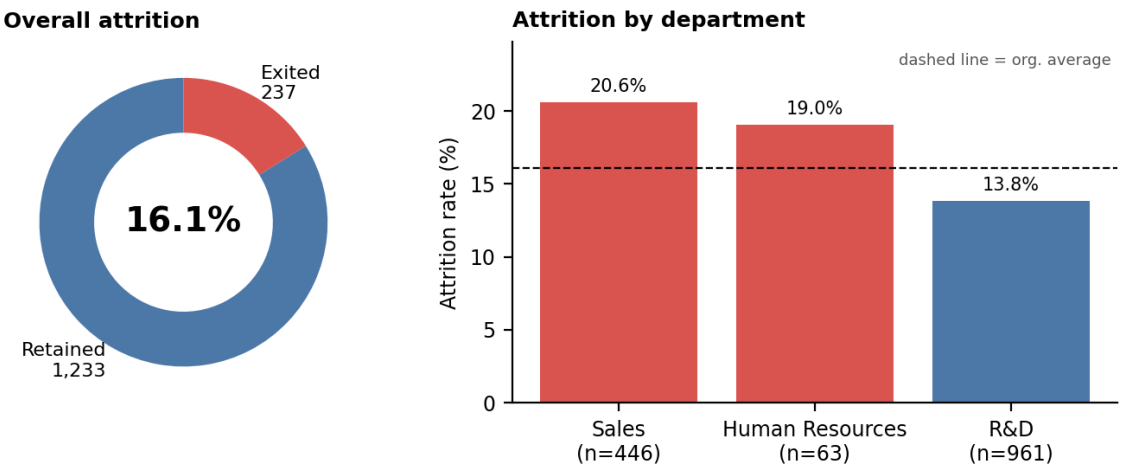


Figure 1. Overall attrition and attrition by department. Red bars exceed the organizational average (dashed line).

3.2 Job roles

Attrition varies far more by role than by department (chi-square p = <0.001). Sales Representatives leave at 39.8%, about 2.5 times the baseline, followed by Laboratory Technicians (23.9%), Human Resources staff (23.1%), Sales Executives (17.5%) and Research Scientists (16.1%). By contrast, Research Directors (2.5%), Managers (4.9%), and Healthcare Representatives and Manufacturing Directors (6.9% each) are very stable. Laboratory Technicians, Sales Executives and Sales Representatives together account for 152 of the 237 exits (64%).

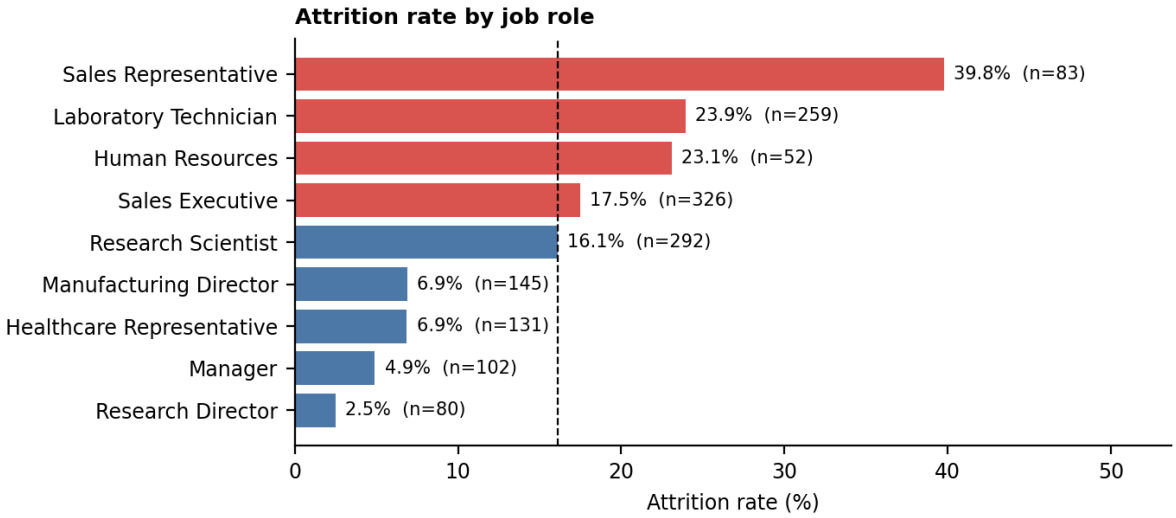


Figure 2. Attrition rate by job role (n = headcount). The role pattern follows seniority: junior, front-line roles have the highest attrition.

3.3 Demographic patterns

Age is one of the strongest demographic signals (p = <0.001). Employees aged 18-24 leave at 39.2% and those aged 25-34 at 20.2%, while attrition falls to about 10% for ages 35-54 and rises slightly to 15.9% at 55+ (a small group of 69, likely reflecting retirement). Employees under 35 are 44% of headcount but 63% of exits. Single employees leave at 25.5% compared with 12.5% for married and 10.1% for divorced employees (p = <0.001). By contrast, **gender** (female 14.8%, male 17.0%; p = 0.291) and **education level** (p = 0.546) show no statistically significant difference. Among fields of study, Human Resources (25.9%, n = 27), Marketing (22.0%) and Technical Degree (24.2%) are higher than Life Sciences (14.7%) and Medical (13.6%) (p = 0.007).

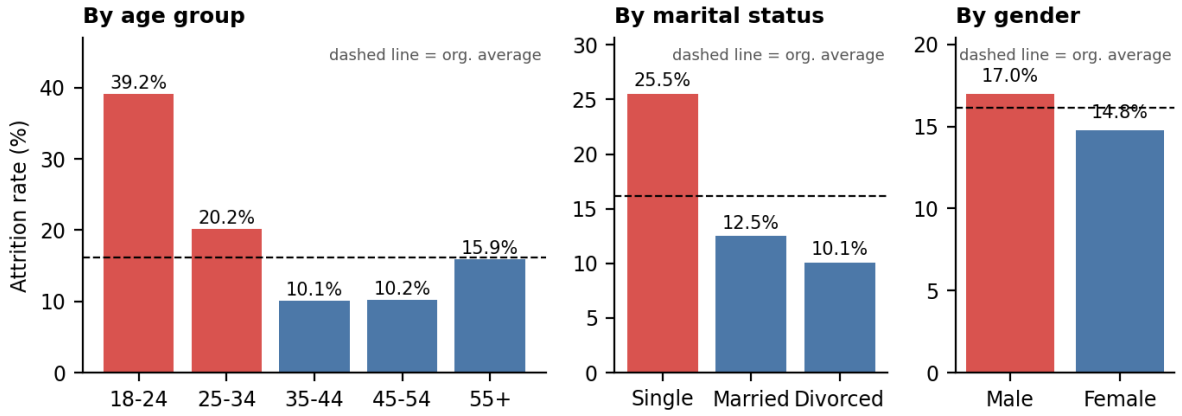


Figure 3. Attrition by age group, marital status and gender.

3.4 Tenure and career stage

Attrition is steepest at the start of the employment relationship. Employees with one year or less at the company leave at 34.9%, falling to 18.4% at 2-3 years, 13.1% at 4-5 years, 12.3% at 6-10 years and 8.1% beyond 10 years (p = <0.001). Grouped by career stage, early-career employees (0-2 years) leave at 29.8%, mid-career (3-9 years) at 12.7% and senior employees (10+ years) at 10.4%. Early-career employees make up 23% of headcount but 43% of all exits, so **early-career employees are leaving more frequently than experienced ones**.

Promotion stagnation is not supported as a driver in this data. Attrition is 17.0% for employees promoted within the last year, 17.1% at 2-3 years, 9.4% at 4-6 years and 15.8% at 7+ years, with no consistent trend and no statistically significant difference (p = 0.157).

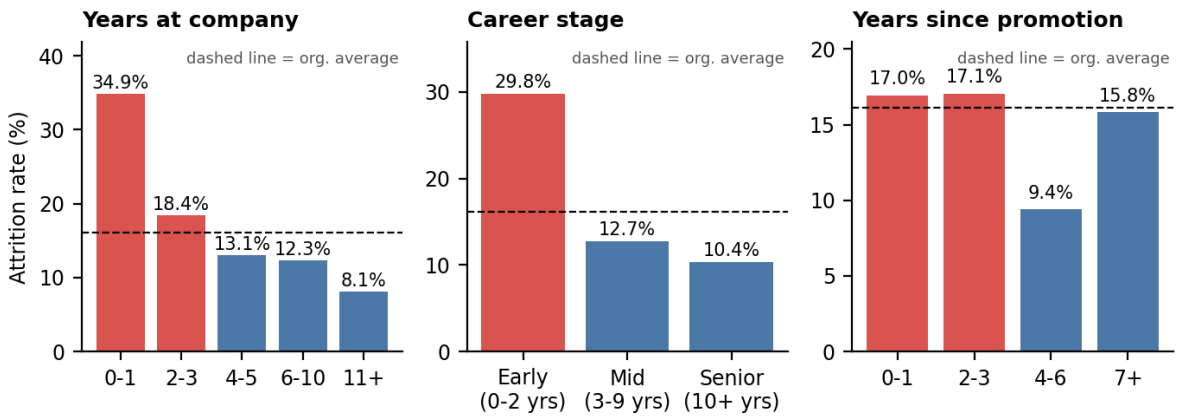


Figure 4. Attrition by tenure band, career stage and years since last promotion.

3.5 Workload and mobility

Workload shows the largest effects in the study. Employees who work overtime leave at 30.5%, roughly three times the 10.4% for those who do not (p = <0.001). Overtime workers are 28% of headcount but 54% of exits. Frequent business travelers leave at 24.9%, compared with 15.0% for rare travelers and 8.0% for non-travelers (p = <0.001). Commute distance has a smaller but visible relationship: attrition is 13.8-14.5% within 10 units of home and 20.0-22.1% beyond 10 (p = 0.010; the dataset does not state the distance unit).

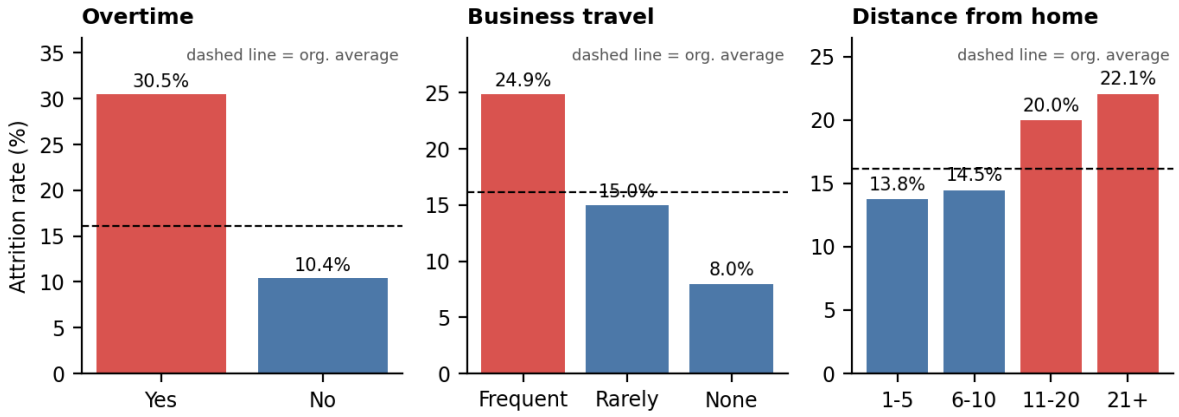


Figure 5. Attrition by overtime, business travel and distance from home.

The two workload factors compound. Among employees who work overtime *and* travel frequently, attrition reaches 41.9% (n = 86), against 4.3% for employees with neither (n = 115). Overtime raises attrition in every department: in Sales from 13.8% to 37.5%, in R&D from 8.6% to 27.3% and in HR from 15.2% to 29.4%. Employees exposed to either overtime or frequent travel are 41% of headcount but 68% of exits, giving a Workload Attrition Index of 2.95.

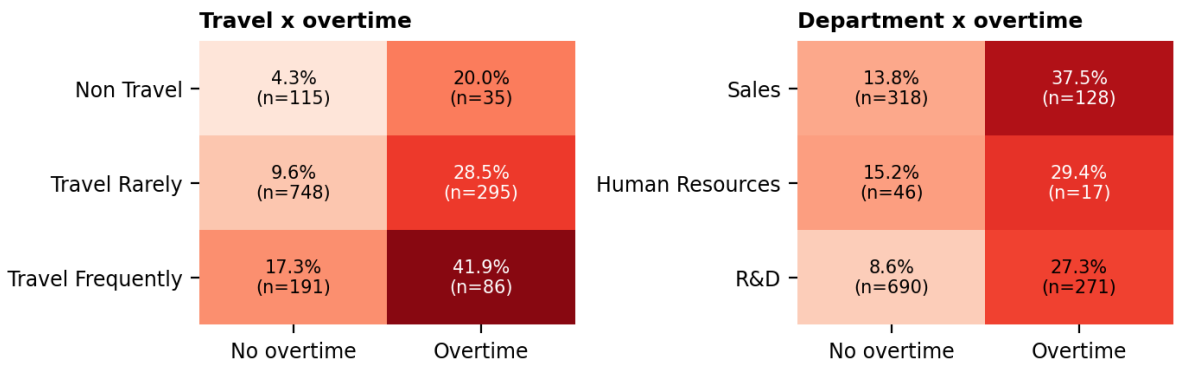


Figure 6. Compounding effect of workload: attrition by travel and overtime, and by department and overtime.

3.6 Do the patterns survive adjustment?

Many of the factors above overlap: junior employees are younger, more likely to be single, paid less and more likely to be in sales roles. To test whether the workload findings are simply a by-product of who is exposed, a logistic regression included all of the variables below at once. Odds ratios above 1 indicate higher odds of attrition.

Factor	Odds ratio
Overtime (yes vs no)	4.37
Travels frequently (vs none)	3.82
Travels rarely (vs none)	1.98
No stock options (vs any)	2.66
Sales Representative role	2.38
Distance from home (per 10 units)	1.43
Single (vs not single)	1.18
Age (per 10 years)	0.77
Job level (per level)	0.74
Years at company (per year)	0.97

Overtime (odds ratio 4.4), frequent travel (3.8 versus no travel), absence of stock options (2.7) and the Sales Representative role (2.4) remain strongly associated with attrition after adjustment. Each additional job level and each additional decade of age is associated with lower odds. Single status falls to a weak association once other factors are controlled (1.18), suggesting that its raw effect largely reflects age and seniority. The linear tenure term is small because the tenure effect is concentrated in the first year or two rather than spread evenly. The model has an in-sample AUC of 0.79; it is used here only to describe associations, not as a validated prediction tool.

4. Risk Hotspot Register

The table consolidates the segments that most clearly exceed the organizational baseline. Segments overlap (for example, entry-level, young and single employees are often the same people), so shares of exits should not be added together.

Segment	Headcount	Exits	Attrition rate	Share of all exits
Overtime + frequent travel	86	36	41.9%	15%
Sales Representatives	83	33	39.8%	14%
Age 18-24	97	38	39.2%	16%
First year or less at company	215	75	34.9%	32%
Overtime workers	416	127	30.5%	54%
Job level 1 (entry)	543	143	26.3%	60%
Single employees	470	120	25.5%	51%
Frequent business travelers	277	69	24.9%	29%
No stock options	631	154	24.4%	65%
Laboratory Technicians	259	62	23.9%	26%
Distance from home 21+	204	45	22.1%	19%
Sales department	446	92	20.6%	39%

Baseline attrition is 16.1%. All segments shown contain at least 30 employees.

5. Discussion and Recommendations

The central finding is that attrition is not evenly spread. It is concentrated among junior, front-line, early-tenure employees who carry heavy workload, and it is relatively insensitive to gender, education level and promotion timing. That points HR toward targeted action rather than company-wide programs. The recommendations below are ordered by the size of the evidence.

Priority	Recommendation	Evidence	Metric to track
1	**Review overtime and travel load.** Audit sustained overtime, especially in Sales and in laboratory roles, and set review triggers for employees who combine overtime with frequent travel.	Overtime 30.5% vs 10.4%; overtime + frequent travel 41.9%; OR about 4	Attrition among overtime workers; Workload Attrition Index
2	**Strengthen the first two years.** Structured onboarding, 30/60/90-day check-ins, buddy or mentor assignment and an early-tenure manager conversation.	34.9% in year 1; 43% of exits from first two years	Early-tenure attrition (0-2 yrs)
3	**Diagnose the Sales Representative role.** Examine compensation structure, targets, territory and overtime expectations for this role before changing broader sales policy.	39.8% attrition; OR 2.4 after adjustment	Role attrition rate
4	**Pilot equity and career-path visibility for entry-level staff.** Employees with no stock options leave at 24.4%, versus 7.6-9.4% for those with option levels 1-2. Entry-level (job level 1) staff leave at 26.3%.	OR 2.7 for no stock options; level 1 = 60% of exits	Attrition at job level 1; stock option coverage
5	**Offer commute flexibility.** Consider hybrid arrangements or commute support for employees who live farther from the workplace.	20.0-22.1% beyond 10 units vs about 14% within	Attrition by distance band
6	**Institutionalize monitoring and collect better data.** Run the dashboard quarterly and add exit dates and exit-reason codes to enable the next phase.	Dataset has no exit date or reason	Quarterly KPI refresh

What not to prioritize: gender-specific, education-specific or promotion-timing programs are not supported by this dataset, because those factors show no statistically significant relationship with attrition here.

Prediction next: the segments identified here are the right starting population for a predictive model, with the added caveat that any model should be validated on held-out data and audited for fairness.

6. Limitations

- **Association, not causation.** Overtime and travel may be a symptom (for example, of understaffing or role design) rather than a direct cause of exit.
- **Snapshot data.** There are no exit dates or exit reasons, so timing and motivation cannot be analyzed.
- **Small groups.** Some segments (Human Resources n = 63, Research Directors n = 80, HR education field n = 27, overtime Sales Representatives n = 24) produce unstable rates and should be read with caution.
- **Multiple comparisons.** Many segments were examined, so borderline results (such as the distance effect) deserve confirmation before major decisions.
- **Model fit.** The logistic model was evaluated on the same data used to fit it and is descriptive only.
- **Dataset provenance.** The role taxonomy (for example Laboratory Technician, Healthcare Representative, Manufacturing Director) differs from what is typical of a cybersecurity company, so the file should be treated as an anonymized or illustrative workforce dataset supplied for this project. Findings describe this dataset.

7. Companion Dashboard

A Streamlit dashboard accompanies this paper and reproduces every analysis interactively. It provides four modules: an attrition overview (overall rate and retained versus exited distribution), department and role heatmaps with automatic hotspot flagging, a demographic explorer with age, gender and education filters, and a tenure and workload analysis. Users can filter by department, job role, tenure range, overtime and business travel, and adjust the early-tenure cutoff.

8. Conclusion

Attrition at Palo Alto Networks, at 16.1% overall, is concentrated rather than general. The strongest and most consistent patterns point to workload (overtime and frequent travel), early tenure and youth, entry-level and front-line roles (particularly Sales Representatives and Laboratory Technicians), and the absence of stock options. Gender, education level and promotion gap do not distinguish leavers from stayers in this data. These diagnostic findings let HR leaders replace intuition-led, company-wide responses with targeted, measurable interventions, and they provide the segments and variables on which a predictive and causal next phase can be built.